

KẾ HOẠCH SỐ HÓA TOÀN BỘ TINH HOA TAM THỨC

Lục Nhâm – Kỳ Môn Độn Giáp – Thái Ất Thần Số

PHẦN 4: KẾ HOẠCH TRIỂN KHAI TOÀN DIỆN

Architecture, Tech Stack, Development Phases, UI/UX, Business Model, Risks & Roadmap

Phiên bản chi tiết nhất – Dành cho CTO, Project Manager & Investors

MỤC LỤC

1. System Architecture Tổng thể
2. Tech Stack Chi tiết
3. Development Phases & Timeline
4. UI/UX Specifications & User Flows
5. Business Model & Go-to-Market
6. Risk Register & Mitigations
7. Legal & Ethical Framework
8. Scaling & Future Roadmap

1. SYSTEM ARCHITECTURE TỔNG THỂ

High-level: Frontend (Next.js/React – interactive charts, dashboards) → Backend (Python FastAPI – calc engines, rule engine, RAG) → AI Layer (RAG + LLM API) → Data Layer (PostgreSQL + Vector DB + Redis). Calculation engines adapt open-source (qimen, kintaiyi, kinliuren) hoặc implement từ ALGORITHM.md. Knowledge Base: Curated patterns + classical excerpts trong Vector DB cho RAG.

2. TECH STACK CHI TIẾT

- Frontend: Next.js 14, React, TypeScript, Tailwind, D3.js / SVG cho interactive 9-cung charts (zoom, rotate, hover details).
- Backend: Python 3.11+, FastAPI, Pydantic. Adapt/open-source engines cho calc. Rule engine (JSON/YAML + Python logic).
- AI: RAG (LangChain / LlamaIndex + Vector DB như Chroma/Pinecone). LLM API (Grok hoặc tương đương) với custom prompts.
- Data: PostgreSQL (users, queries, history), Vector DB (knowledge), Redis (cache).
- Calendar: Thư viện chuẩn cho Ganzhi/Jieqi (hoặc implement từ công thức thiên văn).
- Deployment: Docker, Kubernetes (scale), Vercel/Render cho frontend, AWS/GCP cho backend.

3. DEVELOPMENT PHASES & TIMELINE (12-18 THÁNG MVP → FULL)

Phase 1 (M1-M3): Research & Prototype – Validate algos với master, implement Qi Men core (code sẵn nhiều nhất), basic UI chart, calendar engine.

Phase 2 (M4-M7): Core Engine – Full 3 hệ thống + cross-validation, rule-based patterns, basic RAG.

Phase 3 (M8-M10): AI & Knowledge – Curate KB, prompt engineering, expert review, multi-level output.

Phase 4 (M11-M13): UI/UX & Features – Interactive visuals, scenario tools, reports, personalization.

Phase 5 (M14-M16): Beta & Launch – Test với practitioners + general users, feedback, marketing.

Phase 6 (M17+): Scale – Enterprise features, API, mobile, education hub, community.

4. UI/UX SPECIFICATIONS & USER FLOWS

Key Screens: Landing (value prop + demo), Query Input (date/time + query type + optional birth data), Chart View (interactive 9-cung + metadata), Interpretation (multi-level + citations), Comparison (scenario side-by-side), Dashboard (personal/enterprise), Learning Hub.

Principles: Progressive disclosure (simple → expert), Beautiful visuals (color-coded cát/hung), Mobile-first / PWA, Accessibility (WCAG), Dark mode.

User Flow chính: User chọn 'Timing cho launch' → Nhập proposed dates + context → Generate → View charts + AI summary → Drill-down expert mode → Export report.

5. BUSINESS MODEL & GO-TO-MARKET

Freemium: Basic charts free (limited queries/month). Premium: Full AI + reports + history + priority (₫199k-499k/tháng hoặc annual).

Enterprise: Custom integration, team seats, SLA, expert review add-on (pricing custom, từ \$500+/tháng).

Education: Courses, certification lite, workshop (B2C + B2B).

GTM: Content marketing (blog, YouTube tutorials về Tam Thức), partnerships với master/consultants, SEO cho 'xem ngày tốt', 'Kỳ Môn online', launch tại VN + expand regional (Singapore, Taiwan, US Vietnamese communities).

6. RISK REGISTER & MITIGATIONS

- Accuracy of calculations: Mitigate – Use validated open-source + expert review + unit tests vs classical examples.
- AI hallucination / misinterpretation: Mitigate – Strong RAG + source trace + human audit + disclaimers.
- Cultural sensitivity / backlash: Mitigate – Frame as 'cultural heritage decision support', partner with cultural orgs, strong education focus.
- Legal/Regulatory: Mitigate – Clear disclaimers, position as analytical tool not fortune-telling, consult lawyer for VN/ target markets.
- Over-reliance by users: Mitigate – Education hub, 'use alongside rational analysis', confidence scores.

7. LEGAL & ETHICAL FRAMEWORK

Disclaimers (prominent in all outputs): 'Inspired by traditional methods for reflection and cultural insight. Not guaranteed predictions. Use alongside rational analysis, data, and professional advice.'

Positioning: 'Decision support system rooted in Chinese strategic heritage and symbolic analysis' – for business/strategy consulting.

Privacy: Encrypted personal data (birth info), GDPR-like compliance, clear consent.

Ethics: No over-promising, promote critical thinking, human oversight on core content.

8. SCALING & FUTURE ROADMAP

- API public cho integrations (calendar apps, BI tools).
- Mobile native apps (iOS/Android).
- Education platform (interactive courses, community).
- White-label cho consultants/firms.

- Expand languages (EN, ZH) & markets.
- Long-term: Research partnerships, contribute back to open knowledge on Tam Thức.

TÀI LIỆU THAM KHẢO & NGUỒN KỸ THUẬT

Open-source: anthonylee1994/qimen, kentang2017/kintaiyi & kinliuren, banderzhm/ZhouYiLab. Sách: Joey Yap, Ho Peng Yoke. Các nguồn từ web search trước đó.